APPENDIX

B

MODEL QUESTION PAPERS

MODEL QUESTION PAPER - 1

Time : 3 Hours **Max Marks : 8**

Instruction : Answer all Sections.

SECTION - A

I. Answer any Eight questions. Each question carries 2 Marks. **(8 × 2 = 16**

1. Define Artificial Intelligence.
2. List any two advantages of Artificial Intelligence.
3. What is an Intelligent Agent?
4. Give two examples of goal-based agents.
5. Define Breadth-First Search.
6. Mention two differences between DFS and BFS.
7. What is a Heuristic Function in A* Search?
8. Write any two features of Minimax Search.
9. Define Knowledge-Based Agent.
10. What is Fuzzy Logic?

SECTION - B

II. Answer any Four questions. Each question carries 6 marks. **(4 × 6 = 2**

11. Explain the characteristics of an Intelligent Agent.
12. Discuss the PEAS representation of the Wumpus World with a suitable example.
13. Describe the working of Breadth-First Search algorithm.
14. Write short notes on Constraint Satisfaction Problems (CSPs).
15. Explain the difference between Propositional Logic and Predicate Logic.
16. Write a note on Forward and Backward Chaining with examples.

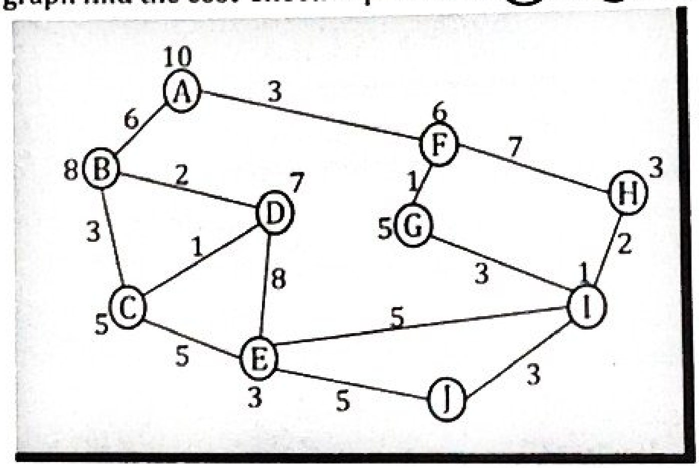
SECTION - C

III. Answer any Five questions. Each question carries 8 marks. **(5 × 8 = 40)**

17. Discuss the structure of Intelligent Agents with a neat diagram.
18. Explain the types of Artificial Intelligence with examples.
19. Describe in detail the working of the A* Search algorithm with an example.
20. (a) Explain the concept of Adversarial Search.

    (b) Write the algorithm for Alpha-Beta Pruning.
21. (a) Explain the process of Unification and Resolution in First-Order Logic with suitable examples.

    (b) For the following graph find the cost-effective path form A to J using A* algorithm.



- Numbers on the edges are distances between the nodes.
- Numbers on the nodes are heuristic value

22. (a) Describe the Blocks World problem in Planning and explain its importance in AI.

    (b) For the following sentences using resolution prove that " John likes peanuts".
    1. John likes all kinds of food.
    2. Apples are food.
    3. Chicken is food.
    4. Anything anyone eats and isn't killed by is food.
    5. Bill eats peanuts and is still alive.
    6. Sue eats everything Bill eats.
23. Explain the architecture and working of an Expert System.

MODEL QUESTION PAPER - 2

Time : 3 Hours **Max Marks : 80**

Instruction : Answer all Sections.

SECTION - A

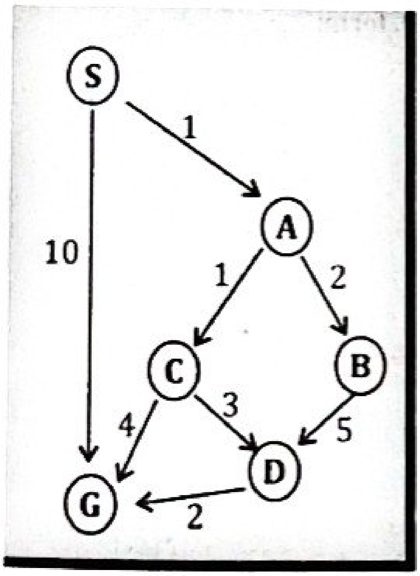
I. Answer any Eight questions. Each question carries 2 Marks. (8 × 2 = 16)

1. Define an Agent Program.
2. Write any two examples of rational agents.
3. What is the difference between agent function and agent program?
4. Define Depth-First Search.
5. List any two limitations of Breadth-First Search.
6. What is the purpose of an Evaluation Function?
7. Define Minimax Algorithm.
8. Briefly explain Turing Test.
9. Define Forward Chaining.
10. Write any two advantages of Fuzzy Logic.

SECTION - B

II. Answer any Four questions. Each question carries 6 marks. (4 × 6 = 24)

11. Explain the different types of environments in Artificial Intelligence.
12. Using A* algorithm find the most cost effective path from S to G.



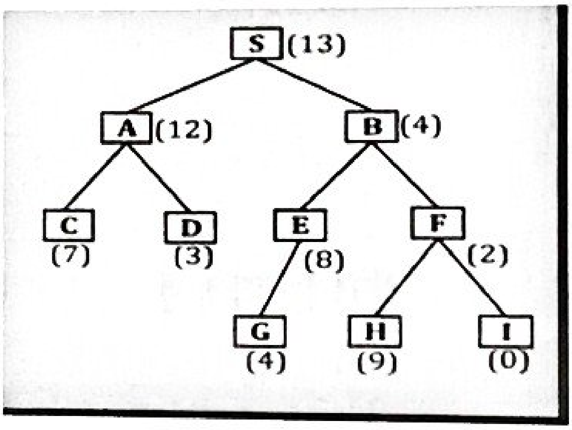
Heuristic	Values
S	5
A	3
B	4
C	2
D	6
G	0

13. Explain the Iterative Deepening Search algorithm with an example.
14. Describe the working of the AO* Search algorithm.
15. Explain the characteristics and structure of First-Order Predicate Logic.
16. Discuss the components and applications of a Truth Maintenance System (TMS).

SECTION - C

III. Answer any Five questions. Each question carries 8 marks. **(5 × 8 = 40)**

17. Explain in detail the architecture of Intelligent Agents and the need for agent design.
18. Discuss the different types of Agent Programs with suitable examples.
19. Describe Best-First Search algorithm with steps and suitable example.
20. (a) Explain the working of Minimax Algorithm.
    (b) Discuss the advantages and disadvantages of Alpha-Beta Pruning.
21. (a) Describe the process of Inference in Artificial Intelligence with Forward and Backward Chaining examples.
    (b) Solve the below given problem using BSF Algorithm



22. Explain the STRIPS representation and its role in Planning.
23. (a) Discuss in detail the features and applications of Machine Learning.
    (b) Convert the following English sentences into FOPL form:
        i) "All students are hardworking."
        ii) "Some teachers are researchers."
        iii) "Every cat chases a mouse."
        iv) "Ravi is a doctor and he treats patients."

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MODEL QUESTION PAPER - 3

Time : 3 Hours **Max Marks : 80**

Instruction : Answer all Sections.

SECTION - A

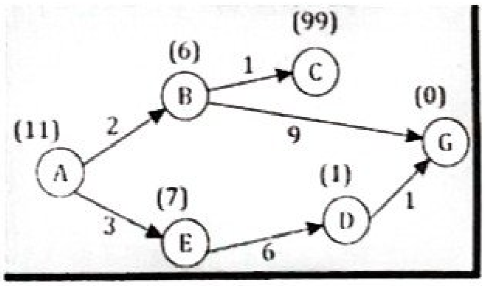
I. Answer any Eight questions. Each question carries 2 Marks. (8 × 2 = 16)

1. Define Rationality in Artificial Intelligence.
2. List any two types of environments in AI.
3. What are Reflex Agents?
4. Mention two disadvantages of Artificial Intelligence.
5. What is the difference between Informed and Uninformed Search?
6. Define Heuristic Search.
7. List any two features of Genetic Algorithms.
8. Define Propositional Logic.
9. What are Quantifiers in Predicate Logic?
10. Write any two applications of Expert Systems.

SECTION - B

II. Answer any Four questions. Each question carries 6 marks. (4 × 6 = 24)

11. Explain the PEAS (Performance, Environment, Actuators, Sensors) framework in Artificial Intelligence.
12. Describe the working of Simple Reflex Agents with neat diagram.
13. Explain the Depth-First Search algorithm with its advantages and disadvantages.
14. Using 'A' algorithm find the optimal path for the following graph. 'A' is the starting node and 'G' is the goal node.



15. Explain the concept of Resolution in First-Order Predicate Logic.
16. Discuss the different types of Planning in Artificial Intelligence.

SECTION - C

III. Answer any Five questions. Each question carries 8 marks. (5 × 8 = 40)

17. (a) Explain in detail the various types of Intelligent Agents with examples.

    (b) Convert the following statements into clausal form:

a) $\forall x\,(P(x) \rightarrow Q(x))$ b) $\exists x\,(P(x) \land R(x))$

18. (a) Discuss the role of Knowledge-Based Agents and how they perform actions in an environment.

(b) Construct truth tables for the following expressions:

a) $\neg P \lor Q$ b) $(P \land Q) \rightarrow R$ c) $P \leftrightarrow (\neg Q \lor R)$

19. Explain in detail the working of the AO* Search algorithm with suitable example.

20. (a) Describe the principles of Game Theory in AI.

(b) Explain the working of Adversarial Search with a Two-Player Zero-Sum Game.

21. Discuss the process of Unification and Lifting with examples.

22. (a) Explain the Blocks World problem and STRIPS in detail.

(b) Write short notes on Genetic Algorithms and their applications in AI.

23. Describe the applications and limitations of Robotics in Artificial Intelligence.

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MODEL QUESTION PAPER - 4

Time : 3 Hours | Max Marks : 80

Instruction : Answer all Sections.

SECTION - A

I. Answer any Eight questions. Each question carries 2 Marks. (8 × 2 = 16)

1. What is the meaning of Environment in Artificial Intelligence?
2. Define the term Agent Architecture.
3. What are Learning Agents?
4. List any two limitations of Intelligent Agents.
5. Define Iterative Deepening Search.
6. Write any two advantages of A* Search Algorithm.
7. Define Constraint Satisfaction Problem (CSP).
8. What is Knowledge Representation?
9. Define Non-Monotonic Reasoning.
10. Mention any two applications of Natural Language Processing (NLP).

SECTION - B

II. Answer any Four questions. Each question carries 6 marks. (4 × 6 = 24)

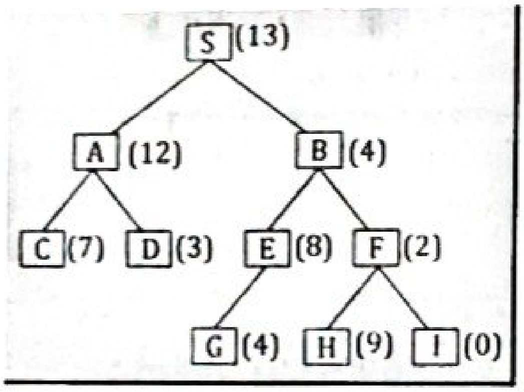
11. Explain the different types of Agent Programs in Artificial Intelligence.
12. Discuss the structure and working of a Model-Based Reflex Agent.
13. Explain the concept of Informed Search and its importance in AI.
14. Describe the working of Minimax Algorithm with suitable example.
15. Explain Forward Chaining and Backward Chaining with proper comparison.
16. Write a note on Probabilistic Reasoning and its applications.

SECTION - C

III. Answer any Five questions. Each question carries 8 marks. (5 × 8 = 40)

17. Explain the architecture and components of Intelligent Agents with neat diagram.
18. (a) Discuss in detail the working and characteristics of Learning Agents.
    (b) Apply quantifiers to express the following:
        i) "Not all birds can fly." ii) "There exists a person who likes coffee."
19. (a) Explain the A* Search algorithm with its evaluation function and step-by-step example.
    (b) There exists a teacher who guides every student. Represent this by introducing a suitable constant.
20. (a) Discuss the concept of Genetic Algorithms in detail.
    (b) Write the general algorithm and applications of Genetic Algorithms.

21. Explain Propositional Logic and First-Order Logic with examples.
22. Describe in detail the concept of Fuzzy Logic and its applications in real life.
23. (a) Explain the working of Deep Neural Networks and their major components.
    (b) Consider the following tree, find the path to reach the goal node I from S using Best First Search Algorithm.



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MODEL QUESTION PAPER - 5

Time : 3 Hours **Max Marks : 80**

Instruction : Answer all Sections.

SECTION - A

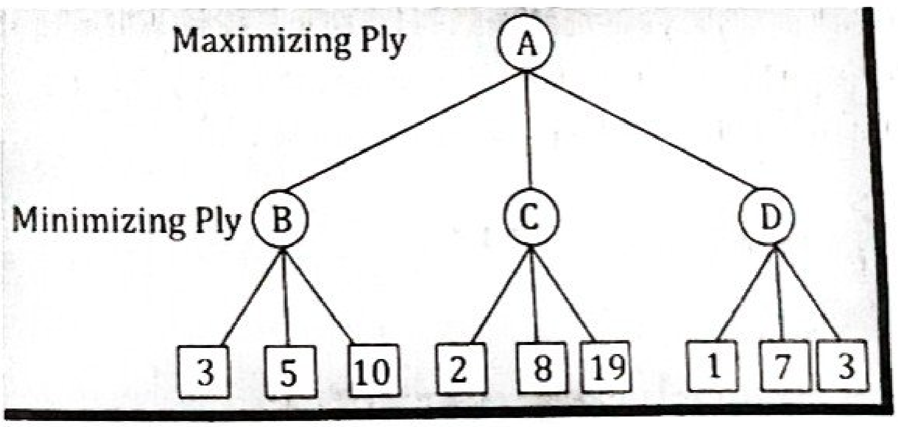
I. Answer any Eight questions. Each question carries 2 Marks. (8 × 2 = 16)

1. Define Artificial Intelligence according to John McCarthy.
2. List any two applications of AI in daily life.
3. What is a Utility-Based Agent?
4. Write any two advantages of Model-Based Reflex Agents.
5. Define Adversarial Search.
6. What is Alpha-Beta Pruning?
7. Define a Genetic Algorithm.
8. What are Quantifiers in Predicate Logic?
9. Write any two advantages of using Fuzzy Logic.
10. What is Deep Learning ?

SECTION - B

II. Answer any Four questions. Each question carries 6 marks. (4 × 6 = 24)

11. Explain the characteristics and need for Rational Agents in AI.
12. Solve the following game tree using alpha beta pruning



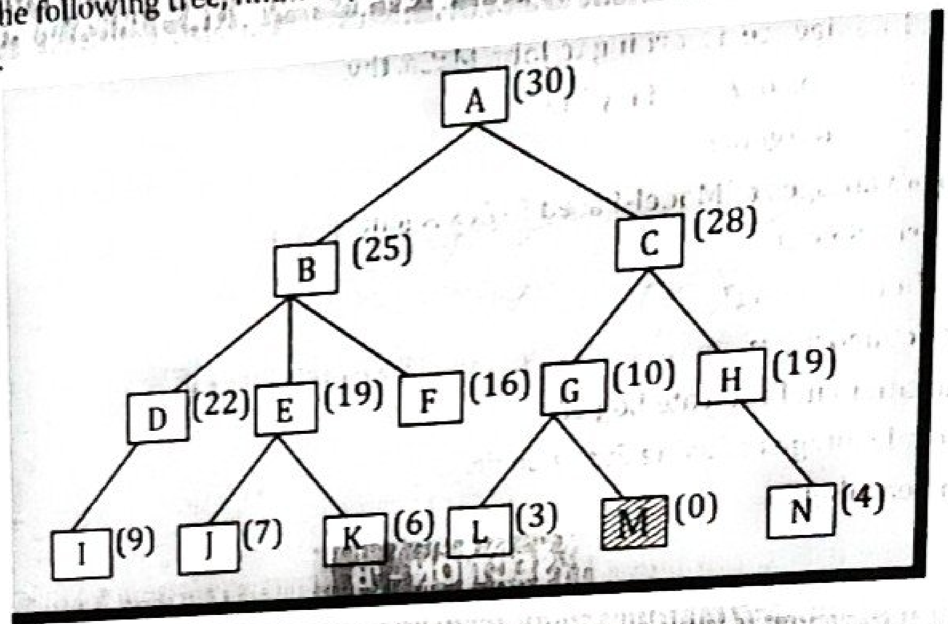
13. Explain the working of the Best-First Search algorithm.
14. Write short notes on Evolutionary Search Techniques in Artificial Intelligence.
15. (a) Explain the concept and process of Truth Maintenance Systems (TMS).

    (b) Explain Tic-Tac-Toe Game Using Minimax Search
16. What is Backtracking? Explain 8-Queens problem with a neat diagram.

SECTION - C

III. Answer any Five questions. Each question carries 8 marks. (5 × 8

17. Explain the types of Environments in Artificial Intelligence with examples.
18. (a) Discuss the structure and working of Knowledge-Based Agents with a neat diagram.
    (b) Consider the following tree, find the path to reach the goal node M from A using Best First Sear Algorithm.



19. Explain the AO* Search algorithm in detail with its features and advantages.
20. (a) Describe the role of Evolutionary Search in solving optimization problems.
    (b) Explain the applications of Genetic Algorithms in Artificial Intelligence.
21. Explain the concept of Non-Monotonic Reasoning and its relation with Truth Maintenance Systems.
22. (a) Describe in detail the applications of Robotics and Computer Vision in modern AI systems.
    (b) Describe the importance and applications of Planning in AI.
23. Explain the architecture, features, and applications of Expert Systems.

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